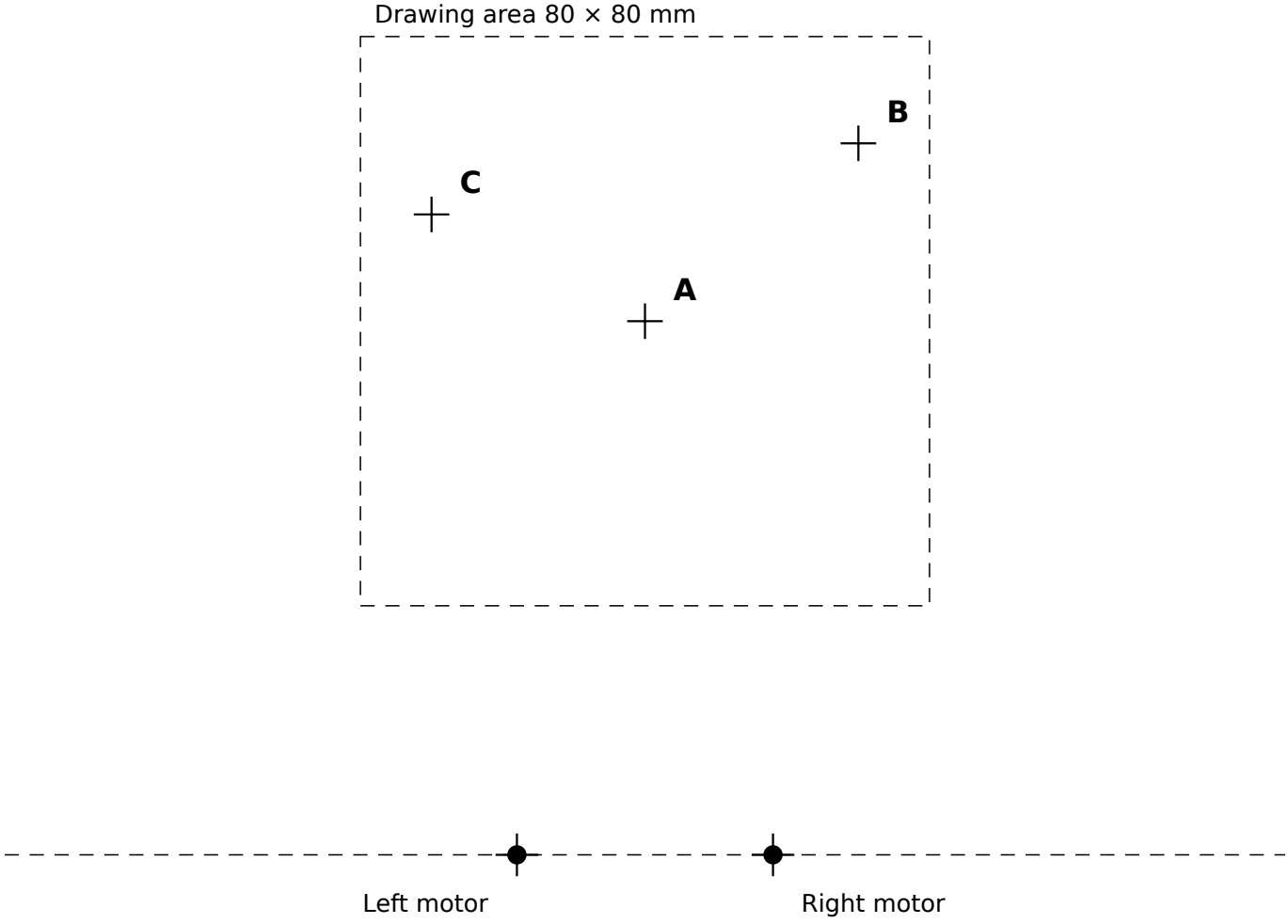
Worksheet: find the motor angles with a compass and protractor

PlotBot from above, scale 1:1. Upper arm 65 mm, forearm 90 mm, motors 36 mm apart.

1. Set the compass to 65 mm (upper arm). Draw a circle around each motor.
2. Set the compass to 90 mm (forearm). Draw a circle around target A.
3. The new circle crosses each motor circle twice. Mark the outer crossing point on each side: that's the elbow.
4. Join them up with a ruler: motor – elbow – target.
5. Measure the angle at each motor with zero pointing right. Write it in the table. Then do the same for B and C.

Target	Left motor	Right motor
A		
B		
C